

TWO PHASE INFLOW PERFORMANCE RELATIONSHIP (IPR) continued

For a two phase (saturated oil) reservoir, if the Vogel's equation is used for constructing the IPR curve, the model constant $q_{\max}$ can be determined by:

$$q_{\max} = \frac{q_t}{\left[1 - 0.2 \left(\frac{P_{wf}}{P} \right) - 0.8 \left(\frac{P_{wf}}{P} \right)^2 \right]}$$

The productivity index at and above bubble point, if desired, can be estimated by

$$J^* = \frac{1.8q_{\max}}{\bar{P}}$$

If Fetkovich's equation is used two test points are required for determining the value of the two model constants, that is:

$$n = \frac{\log \left(\frac{q_1}{q_2} \right)}{\log \left(\frac{P^2 - P_{wf1}^2}{P^2 - P_{wf2}^2} \right)}$$

And

$$c = \frac{q_1}{\left(P^2 - P_{wf1}^2 \right)^n}$$

Where q_1 and q_2 are the tested production rates at tested flowing bottom hole pressure P_{wf1} and P_{wf2} respectively.

Question 6

Construct IPR of a well in a saturated oil reservoir using both Vogel's and Fetkovich's equations. The following data are given

Reservoir pressure $\bar{P} = 3000\text{psia}$

Tested flowing bottom hole pressure in well A $P_{wf1} = 2000\text{psia}$

Tested production rate from well A $= q_1 = 500\text{ stb/day}$

Tested flowing bottom hole pressure in well B $P_{wf2} = 1000\text{psia}$

Tested production rate from well B = $q_2 = 800$ stb/day

SOLUTION

Vogel's equation

$$q_{\max} = \frac{q_i}{\left[1 - 0.2 \left(\frac{P_{wf}}{P} \right) - 0.8 \left(\frac{P_{wf}}{P} \right)^2 \right]}$$

$$= \frac{500}{1 - 0.2 \left(\frac{2000}{3000} \right) - 0.8 \left(\frac{2000}{3000} \right)^2}$$

=978 stb/day

$$J^* = \frac{1.8q_{\max}}{\bar{P}}$$

$$J^* = 1.8 \cdot 978 / 3000$$

$J^* =$

The calculated points are

P_{wf} (psia)	q (stb/day)
0	978
500	924
1000	826
1500	685
2000	500
2500	272
3000	0

Fetkovich's equation

$$n = \frac{\log\left(\frac{q_1}{q_2}\right)}{\log\left(\frac{P^2 - P_{wf1}^2}{P^2 - P_{wf2}^2}\right)}$$

And

$$c = \frac{q_1}{\left(P^2 - P_{wf1}^2\right)^n}$$

Where q_1 and q_2 are the tested production rates at tested flowing bottom hole pressure P_{wf1} and P_{wf2} respectively.

$$n = \frac{\log\left(\frac{500}{800}\right)}{\log\left(\frac{3000^2 - 2000^2}{3000^2 - 1000^2}\right)}$$

$$n = 1.0$$

$$c = \frac{500}{\left(3000^2 - 2000^2\right)^{1.0}}$$

Calculated points are

P_{wf} (psia)	q (stb/dat)
0	900
500	875
1000	800
1500	675
2000	500
2500	275
3000	0

This indicates that Fetkovich's equation with two constant catches more than Vogel's equation.

Tubing Performance Relation (TPR)

The pressure drop required to lift reservoir fluid to the surface at a given rate is one of the major factors affecting well deliverability.

Other names of TPR include

- Vertical lift performance
- Wellbore flow relationship
- Outflow performance relationship

Up to 80% of total pressure loss may occur in the tubing, part of the tubing include completion equipment e.g. subsurface flow control devices. Additional, tubing string may be composed of multiple tubing diameters. The pressure drop is a function of the mechanical configuration of the wellbore, the properties of the fluids, and the production rates.

Procedures for the determination of the Tubing performance Relationship (TPR)

- i. The well head or bottom hole flowing pressure at a given rate is determined.
- ii. The pressure drop along the tubing can be calculated by charts or correlation.
- iii. The resulting flowing pressure at the other end of the tubing can be determined
- iv. The resulting relationship between bottom hole flowing pressure and production rate is called Tubing Performance Relationship. (TPR), and it is valid only for the specified well head pressure.

The interception of the IPR and TPR curves when plotted on the same graph will give one the stabilized flow rate (point of Natural flow). Since at the specific rate when these two pressures are equal the flow system is in equilibrium and the flow is stable.

Pressure loss estimation

- a. Single phase liquid flow: This type of fluid flow is generally of minor interest to the petroleum engineer, except for the cases of water supply or injection wells.
- b. Single phase vapour flow: for dry gas wells there are several correlations to calculate pressure drop in single phase gas flow, they include:
 - Average temperature and compressibility method.
 - Original Cullender and Smith method.
 - Modified Cullender and Smith method.
- c. Multiple flow: pressure drop in multiple flow is more complex than that of a single phase flow because parameters such as velocity, frictional factor, density, and the fraction of the vapour to liquid change as the liquid flows to the surface.

Pressure drop can be determined either by correlations or by gradient curve. Some of the correlations are:

- Dun and Ros (1963)
- Dukler (1964)
- Orkiszewski (1967)
- Hageborn and Brown (1967)
- Beggs and Brill (1973)
- Mukherjee and Brill (1981)

Factors affecting vertical lift

- Flowing tubing pressure
- Length of tubing
- Friction factors
- Tubing diameter
- Fluid composition (liquid/gas)

Tutorial questions

1a. Construct Inflow Performance Relationship Curve of a vertical well in a saturated oil reservoir using Vogel's equation. The following data are given

Porosity $\phi = 0.17$

Effective horizontal permeability = $k = 7.3$ md

Pay zone thickness $h = 75$ ft.

Reservoir pressure $P = 6,342$ psia.

Bubble point pressure = $P_b = 6500$ psia

Fluid Formation Volume Factor = $B_o = 1.3$ rb/stb

Fluid viscosity = $\mu_o = 1.85$ cp

Total compressibility = $C_t = 0.0000132$ psi⁻¹

Drainage Area: $A = 640$ Acres ($r_e = 2980$ ft)

Well bore radius = $r_w = 0.328$ ft

Skin factor = 0.256

(10 marks)

1b. If the Tubing performance Relationship curve can be represented by

$$q = 2.6 * 10^{-6} * DP_{wf}^2 + 8.7 * 10^{-5} * DP_{wf} + 0.057, \text{ where } P_{wf} \text{ is the flowing bottom hole}$$

pressure in psia. and D is the tubing diameter in inches Calculate the change in flow rate

if the tubing of diameter 5 inches is replaced by tubing of diameter 7.5 inches. (20 marks)

2a. Discuss the methods available for the control and prevention of sand production in petroleum production

2b. What are the roles of the Petroleum Production Engineers.

2c. Write on the major types of Well Completion.

2d. Discuss the principles of Gas Oil Separation Plant.

3a. Write on the methods used in oil field emulsions

3b. What are the methods used in tertiary oil recovery

3c. Discuss the well stimulation operations

3d. Explain the principles of natural gas dehydration.

4a. Discuss the casing-tubing configuration commonly used in petroleum industry

4b. State the work-over operations that warranted remedial cementing

4c. A well with initial production of 4500 bopd with decline rate of 0.08416/year, if $b = 0.73$, calculate:

- i. The production during the 16th year
- ii. The total production from the eleventh to the seventeenth year.
- iii. The gas production during the third year if gas oil ratio is 35200 scf/stb